A Symplectic N-Body Engine for Solar System Dynamics: Validation Against NASA JPL Horizons

Karl Keshavarzi*[1]

[1]Department of Electrical & Computer Engineering
University of Waterloo

February 2026

Abstract

We present a high-performance N-body gravitational simulator for solar system dynamics, integrating 35 bodies (the Sun, 8 planets, Pluto, and 25 natural satellites) over timescales of up to 249 years. The simulator employs a 4th-order Yoshida symplectic integrator, achieving energy conservation of $\Delta E/E \leq 7.4 \times 10^{-13}$ over 100 years and $\leq 1.5 \times 10^{-12}$ over 249 years with a timestep of $\Delta t = 360\,\mathrm{s}$. A timestep convergence study confirms that $\Delta t = 360\,\mathrm{s}$ lies on the floating-point precision floor; further reduction yields no improvement in energy conservation. Positional accuracy is validated against the NASA JPL Horizons DE441 ephemeris, yielding a mean maximum relative error of 0.079% over 249 years across 34 non-Sun bodies. Although the Sun attains a larger maximum relative error of 1.336% under barycentric normalization, its maximum absolute position error is only 238 km with RMS absolute error 159 km, confirming that this is a metric artifact rather than a physically significant trajectory failure. Residual errors are attributed to physics model differences; principally, the absence of post-Newtonian relativistic corrections, solar oblateness J_2, and asteroid perturbations, rather than numerical integration error.

Keywords: N-body simulation, symplectic integration, Yoshida integrator, solar system dynamics, JPL Horizons, gravitational dynamics

1. Introduction

The gravitational N-body problem, that is, computing the trajectories of N mutually interacting masses, is one of the oldest and most fundamental problems in computational physics. Accurate long-duration integration of planetary orbits requires numerical methods that preserve the geometric structure of Hamilton's equations. Non-symplectic integrators exhibit secular energy drift that corrupts orbital elements over extended timescales [2].

This work presents a complete N-body simulation framework for solar system dynamics. The system integrates 35 bodies (the Sun, 8 planets, Pluto, and 25 natural satellites) under pairwise Newtonian gravitation using the 4th-order Yoshida symplectic integrator [1]. We validate the simulator against the NASA JPL Horizons system, which provides state vectors computed from the DE441 planetary

*This work was completed independently and is not affiliated with the University of Waterloo.

ephemeris. DE441 is a comprehensive model incorporating post-Newtonian relativity, solar oblateness, asteroid perturbations from over 300 bodies, and tidal effects [3].

The primary contributions of this work are:

1. Implementation and verification of a 4th-order Yoshida symplectic integrator achieving $\Delta E/E \leq 10^{-12}$ over multi-century timescales.
2. A systematic timestep convergence study identifying three distinct regimes of integrator behavior: a floating-point precision floor, 4th-order convergence, and dynamical under-resolution.
3. Comprehensive positional validation against JPL DE441 for 35 solar system bodies, with quantitative attribution of residual errors to specific missing physics.
4. An automated validation pipeline that fetches JPL Horizons ephemerides and performs comparison at matched cadence, avoiding interpolation in the standard workflow.

2. Theoretical Background

2.1. Equations of Motion

The gravitational acceleration of body i due to all other $N-1$ bodies is given by Newton's law of universal gravitation:

$$\mathbf{a}_i = \sum_{\substack{j=1 \\ j\neq i}}^{N} \frac{G\, m_j\, (\mathbf{r}_j - \mathbf{r}_i)}{\left(|\mathbf{r}_j - \mathbf{r}_i|^2 + \varepsilon^2\right)^{3/2}} \tag{1}$$

where $G = 6.6743 \times 10^{-11}\,\mathrm{m^3\,kg^{-1}\,s^{-2}}$ is the gravitational constant, m_j is the mass of body j, and $\mathbf{r}_i$ and $\mathbf{r}_j$ are position vectors. A softening parameter $\varepsilon = 10^{-9}\,\mathrm{m}$ is included to prevent numerical singularity at zero separation. This value is negligible relative to interplanetary distances and does not measurably affect the dynamics; it serves only as a numerical safeguard. The results are insensitive to the choice of ε provided $\varepsilon \ll \min_{i\neq j} |\mathbf{r}_i - \mathbf{r}_j|$.

2.2. Symplectic Integration

Hamiltonian systems possess a symplectic structure: the phase-space flow preserves the canonical 2-form $\sum_i dp_i \wedge dq_i$. Standard Runge-Kutta methods violate this structure, causing secular energy drift proportional to integration time. Symplectic integrators, by construction, approximately preserve a nearby shadow Hamiltonian $\tilde{H} = H + \mathcal{O}(\Delta t^p)$, bounding energy error over exponentially long timescales in $1/\Delta t$ [2]. The energy error remains oscillatory and does not exhibit secular growth.

2.2.1. Velocity Verlet 2nd-Order Integrator

The Velocity Verlet algorithm [4] is a 2nd-order symplectic method widely used in molecular dynamics:

$$\mathbf{r}(t+\Delta t) = \mathbf{r}(t) + \Delta t\, \mathbf{v}(t) + \tfrac{1}{2}\Delta t^2\, \mathbf{a}(t) \tag{2}$$

$$\mathbf{v}(t+\Delta t) = \mathbf{v}(t) + \tfrac{1}{2}\Delta t\, [\mathbf{a}(t) + \mathbf{a}(t+\Delta t)] \tag{3}$$

This requires storing the previous acceleration and one force evaluation per timestep.

2.2.2. Yoshida 4th-Order Integrator

The Yoshida method [1] constructs a 4th-order symplectic scheme by composing three leapfrog sub-steps with carefully chosen coefficients. Define:

$$w_1 = \frac{1}{2 - 2^{1/3}}, \qquad w_0 = -\frac{2^{1/3}}{2 - 2^{1/3}} \tag{4}$$

The position drift and velocity kick coefficients are:

$$c_1 = c_4 = \frac{w_1}{2}, \qquad d_1 = d_3 = w_1 \tag{5}$$

$$c_2 = c_3 = \frac{w_0 + w_1}{2}, \qquad d_2 = w_0 \tag{6}$$

Each timestep proceeds as four position drifts interleaved with three force evaluations ("kicks"):

$$\mathbf{r} \xrightarrow{c_1} \mathbf{v} \xrightarrow{d_1} \mathbf{r} \xrightarrow{c_2} \mathbf{v} \xrightarrow{d_2} \mathbf{r} \xrightarrow{c_3} \mathbf{v} \xrightarrow{d_3} \mathbf{r} \xrightarrow{c_4} \tag{7}$$

Here $\mathbf{r} \xrightarrow{c_k}$ denotes $\mathbf{r} \leftarrow \mathbf{r} + c_k \Delta t\, \mathbf{v}$ and $\mathbf{v} \xrightarrow{d_k}$ denotes $\mathbf{v} \leftarrow \mathbf{v} + d_k \Delta t\, \mathbf{a}(\mathbf{r})$.

The scheme achieves $\mathcal{O}(\Delta t^4)$ local truncation error

with only 3 force evaluations per step, compared to 4 for classical RK4. The negative coefficient $w_0 < 0$ (equivalently $d_2 < 0$) introduces a backward sub-step that cancels lower-order error terms. The symplectic structure guarantees that energy errors remain bounded and oscillatory, never exhibiting secular growth.

3. Implementation

3.1. Software Architecture

The simulator is implemented in C++23 with a modular object-oriented architecture. An abstract `Integrator` base class defines the integration interface, with `Yoshida` and `Velocity_Verlet` as concrete implementations. An abstract `Force` base class permits extensible force models; the current implementation provides `Gravity` for pairwise Newtonian interactions. The `Simulation` class orchestrates the time-stepping loop, conservation diagnostics, and binary output.

All simulation parameters (timestep Δt, duration, and output interval) are specified in a single `Config.hpp` header as `constexpr` values. This enables compile-time computation of derived quantities such as total steps and output cadence. A `static_assert` enforces that the output interval is an exact multiple of the timestep.

3.2. Data Layout: Structure of Arrays

Particle data is stored in a Structure-of-Arrays (SoA) layout rather than the conventional Array-of-Structures (AoS). All 13 sub-arrays (3 positions, 3 velocities, 3 accelerations, 3 old accelerations, and mass) occupy a single contiguous allocation. Each sub-array is explicitly aligned to the target SIMD width (32 bytes for AVX2, 64 bytes for AVX-512) and padded to a SIMD-element boundary so that every array start is naturally aligned for vectorized loads and stores:

$$\texttt{pos_x}[0..\text{stride}-1], \quad \texttt{pos_y}[0..\text{stride}-1], \quad \ldots$$

where the stride is N rounded up to the nearest multiple of the SIMD register width in elements (e.g., 4 doubles for AVX2). This layout maximizes cache-line utilization during the inner force loop and guarantees aligned SIMD access without runtime checks.

Raw array pointers are qualified with compiler-specific `__restrict__` annotations, informing the compiler that no aliasing occurs between position, velocity, and acceleration arrays. This enables aggressive SIMD auto-vectorization of the inner loops.

3.3. Force Computation

The gravitational force calculation uses $\mathcal{O}(N^2)$ direct pairwise summation. Self-interaction ($i = j$) is eliminated with a branchless floating-point mask:

$$\texttt{mask}_j = \begin{cases} 0.0 & \text{if } i = j \\ 1.0 & \text{otherwise} \end{cases} \tag{8}$$

The force contribution is multiplied by this mask, replacing a conditional branch that would prevent SIMD vectorization of the inner loop. The inner loop is annotated with `#pragma omp simd` with reductions on the acceleration accumulators to enable vectorized execution. For $N \geq 500$, the outer loop is parallelized with OpenMP via `#pragma omp parallel for`.

Note that Newton's third law symmetry ($\mathbf{F}_{ij} = -\mathbf{F}_{ji}$) is not exploited; each body's acceleration is computed independently by summing over all N others. This doubles the floating-point work relative to a symmetric $j > i$ loop but is a deliberate optimization tradeoff: the regular access pattern enables SIMD vectorization of the full inner loop, and the independent per-body accumulation eliminates write conflicts under thread parallelism without requiring atomic operations.

3.4. Initial Conditions

State vectors for all 35 bodies are sourced from the NASA JPL Horizons system at epoch J1950.0 (January 1, 1950, 00:00 UTC). The reference frame is

solar system barycentric (SSB) with the ecliptic reference plane (ICRF). Positions are given in kilometers and velocities in $\mathrm{km\,s^{-1}}$, converted to SI units at initialization. The body catalog comprises the Sun, 8 planets (Mercury through Neptune), Pluto, and 25 natural satellites: the Moon, Phobos, Deimos, the four Galilean moons, Amalthea, 9 Saturnian moons (Mimas through Phoebe), 5 Uranian moons (Ariel through Miranda), Triton, Nereid, and Charon.

The simulation operates in the SSB frame throughout. Because all initial state vectors are sourced from JPL Horizons in this frame, the total momentum of the 35-body subset is not identically zero (the missing solar system bodies carry the complementary momentum). No barycentric correction is applied.

An automated Python pipeline (`jpl_compare.py fetch`) queries the Horizons API, retrieves state vectors at the configured output cadence, and generates both the C++ initial conditions header and the reference ephemeris for validation.

3.5. Build System

The project uses CMake (minimum version 3.25) for cross-platform builds. Compiler optimizations (`-O3 -march=native`) and OpenMP linkage are configured automatically. The build targets GCC, Clang, and MSVC.

4. Validation Methodology

Validation is performed by direct comparison of simulated trajectories against JPL Horizons ephemeris data, supplemented by conservation diagnostics. The pipeline ensures methodological rigor through several design choices.

Epoch-aligned comparison. Both the simulator and the JPL fetch use identical output intervals (`output_hours`), sourced from the same `Config.hpp` file. This avoids interpolation in the standard workflow, since both output streams sample at the same cadence from the same start epoch.

Relative position error. For each body at each epoch, the relative position error is defined as:

$$\epsilon_{\mathrm{rel}}(t) = \frac{|\mathbf{r}_{\mathrm{sim}}(t) - \mathbf{r}_{\mathrm{JPL}}(t)|}{|\mathbf{r}_{\mathrm{JPL}}(t)|} \tag{9}$$

where $|\cdot|$ denotes the Euclidean norm. For each body, we report the maximum of $\epsilon_{\mathrm{rel}}(t)$ over all epochs $t > 0$. The *mean maximum relative error* reported in summary tables is the average of this per-body maximum across all bodies in the comparison set. This metric is informative for bodies whose barycentric distance is large relative to the position error, but can be misleading for bodies that remain close to the barycenter (notably the Sun); see Section 5.2 for discussion.

Absolute position error and RMS metrics. To complement the maximum relative error, we also report the absolute position error and RMS statistics:

$$e_{\mathrm{abs}}(t) = |\mathbf{r}_{\mathrm{sim}}(t) - \mathbf{r}_{\mathrm{JPL}}(t)| \tag{10}$$

$$\mathrm{RMS}_{\mathrm{rel}} = \sqrt{\frac{1}{N_t}\sum_t \epsilon_{\mathrm{rel}}(t)^2} \tag{11}$$

$$\mathrm{RMS}_{\mathrm{abs}} = \sqrt{\frac{1}{N_t}\sum_t e_{\mathrm{abs}}(t)^2} \tag{12}$$

These supplemental metrics are particularly important for bodies whose barycentric radius is small, for which the normalization in Eq. (9) can inflate the reported percentage. The RMS metrics characterize typical disagreement over the full integration, in contrast to the maximum which may be dominated by a single epoch.

Energy conservation. Total mechanical energy sampled at regular intervals throughout the integration:

$$T = \sum_{i=1}^{N} \frac{1}{2} m_i \left(v_{x,i}^2 + v_{y,i}^2 + v_{z,i}^2\right) \tag{13}$$

$$U = -G \sum_{i=1}^{N} \sum_{j=i+1}^{N} \frac{m_i\, m_j}{\sqrt{|\mathbf{r}_j - \mathbf{r}_i|^2 + \varepsilon^2}} \tag{14}$$

$$E = T + U \tag{15}$$

Energy conservation is quantified by the peak-to-

peak relative variation:

$$\frac{\Delta E}{E} = \frac{|E_{\max} - E_{\min}|}{|E_0|} \tag{16}$$

where $E_{\max}$ and $E_{\min}$ are the extrema over all sampled epochs, and E_0 is the initial energy. Note that this peak-to-peak measure can differ by up to a factor of 2 from the more commonly reported metric $\max_t |E(t) - E_0|/|E_0|$ [2, 7]. The peak-to-peak definition provides a conservative upper bound on the energy variation.

Angular momentum conservation. Total angular momentum is computed as

$$\mathbf{L} = \sum_{i=1}^{N} m_i \left(\mathbf{r}_i \times \mathbf{v}_i\right) \tag{17}$$

and its conservation is quantified by the maximum vector drift relative to the initial state:

$$\frac{\Delta L}{L} = \frac{\max_t |\mathbf{L}(t) - \mathbf{L}(0)|}{|\mathbf{L}_0|} \tag{18}$$

This vector-based definition catches both magnitude changes and directional rotation of the angular momentum vector, unlike a scalar magnitude comparison which can miss rotational drift. For an isolated Newtonian system, total angular momentum is conserved. The symplectic integrator preserves it to the accuracy of the shadow Hamiltonian; observed deviations are at the level of floating-point arithmetic. Angular momentum conservation provides an independent diagnostic from energy, as it is sensitive to different classes of numerical error.

Linear momentum conservation. Total linear momentum is computed as

$$\mathbf{P} = \sum_{i=1}^{N} m_i \, \mathbf{v}_i \tag{19}$$

and its drift is quantified analogously as a vector difference:

$$\frac{\Delta P}{P} = \frac{\max_t |\mathbf{P}(t) - \mathbf{P}(0)|}{|\mathbf{P}_0|} \tag{20}$$

Because the simulation evolves only the selected 35 bodies under internal pairwise Newtonian gravity, total linear momentum of the simulated system should remain constant. The initial total momentum is generally nonzero because the selected subset does not represent the complete solar system, but this does not imply physical drift in $|\mathbf{P}|$ over time. Observed variation in $\mathbf{P}$ is therefore interpreted as numerical error rather than physical evolution, providing an additional diagnostic of integrator quality.

5. Results

5.1. 100-Year Simulation

Table 1 summarizes results for a 100-year integration (1950–2050) with $\Delta t = 360\,\mathrm{s}$ and the Yoshida integrator. The simulation completes in 56.5 s on a single core.

Table 1: 100-year simulation summary (35 bodies, $\Delta t = 360\,\mathrm{s}$, Yoshida 4th-order).

Metric	**Value**
Mean max. relative error†	0.053%
Worst-case body†	Ganymede (0.290%)
Planet error range	0.0003%–0.173%
Moon error range	0.009%–0.290%
Energy conservation $\Delta E/E$	7.4×10^{-13}
Angular momentum $\Delta L/L$	3.4×10^{-14}
Linear momentum $\Delta P/P$	1.0×10^{-8}
Wall-clock runtime	56.5 s

†Sun excluded; see Section 5.2.

Table 2 presents the per-body breakdown, grouped by dynamical class.

Table 2: Per-body maximum relative position error (%) for the 100-year simulation.

Body	**Error (%)**	**Body**	**Error (%)**
Sun and Planets			
Saturn	0.0003	Venus	0.022
Uranus	0.0003	Sun	0.071
Neptune	0.0004	Mercury	0.173
Jupiter	0.0009		
Mars	0.006		
Pluto	0.010		
Earth	0.013		
Selected Moons			
Iapetus	0.009	Io	0.114
Miranda	0.010	Titan	0.125
Phobos	0.014	Europa	0.175
Triton	0.016	Callisto	0.188
Moon	0.018	Ganymede	0.290

5.2. 249-Year Simulation

Extending the integration to 249 years (1950–2199) tests the long-term stability of the symplectic integrator. Table 3 compares the two simulation durations, and Table 4 presents the per-body breakdown.

Table 3: Comparison of 100-year and 249-year simulations ($\Delta t = 360\,\mathrm{s}$).

Metric	**100 yr**	**249 yr**
Mean max. error†	0.053%	0.079%
Worst body†	Ganymede (0.29%)	Callisto (0.41%)
Worst planet	Mercury (0.17%)	Mercury (0.42%)
$\Delta E/E$	7.4×10^{-13}	1.5×10^{-12}
$\Delta L/L$	3.4×10^{-14}	4.3×10^{-13}
$\Delta P/P$	1.0×10^{-8}	7.0×10^{-8}
Runtime	56.5 s	171.6 s

†Sun excluded; see Section 5.2.

Table 4: Per-body position error (%) for the 249-year simulation.

Body	**Max Rel (%)**	**RMS Rel (%)**
Sun and Planets		
Saturn	0.0007	0.0004
Uranus	0.0007	0.0004
Neptune	0.0009	0.0005
Jupiter	0.002	0.001
Mars	0.015	0.007
Pluto	0.034	0.016
Earth	0.031	0.018
Venus	0.055	0.031
Mercury	0.417	0.160
Sun	1.336	0.049
Selected Moons		
Miranda	0.010	0.006
Triton	0.016	0.011
Umbriel	0.019	0.013
Iapetus	0.022	0.012
Phobos	0.022	0.009
Moon	0.045	0.019
Amalthea	0.050	0.033
Io	0.116	0.077
Nereid	0.149	0.052
Europa	0.184	0.122
Titan	0.184	0.133
Hyperion	0.226	0.153
Ganymede	0.290	0.194
Callisto	0.411	0.246

Several observations follow from these results.

Energy and angular momentum conservation remain bounded. The energy drift increases only from 7.4×10^{-13} to 1.5×10^{-12} when the integration duration grows by a factor of 2.5. This sub-linear growth is characteristic of symplectic integrators, which approximately preserve a shadow Hamiltonian $\tilde{H}$ and exhibit only oscillatory energy variations, never secular drift. Angular momentum conservation follows the same pattern, with $\Delta L/L$ remaining at $\sim 10^{-13}$ over 249 years.

Linear momentum conservation confirms integrator correctness. Because the simulated system is closed under its own pairwise Newtonian interactions, total linear momentum should be exactly conserved up to numerical error, even though $\mathbf{P}_0 \neq 0$. The observed drift of $\Delta P/P \sim 10^{-8}$ is consistent with accumulated floating-point roundoff

over $\sim 2.2 \times 10^7$ timesteps, confirming that the integrator correctly preserves this conservation law. The growth from 1.0×10^{-8} to 7.0×10^{-8} over 2.5× longer integration is consistent with sub-linear roundoff accumulation.

Positional errors grow sub-linearly. The mean error (excluding the Sun) grows from 0.053% to 0.079% over 2.5× the integration time. This sub-linear growth is consistent with slow phase drift accumulation, not exponential divergence. The outer planets (Saturn, Uranus, Neptune) remain below 10^{-3}% even at 249 years, reflecting the dominance of solar gravitation in their dynamics and the absence of significant missing physics at their orbital distances.

The Sun's large relative error is a metric artifact. Although the Sun attains the largest maximum relative error (1.336%) at 249 years, this is a normalization artifact rather than a simulation failure. Because the comparison is performed in solar-system barycentric coordinates and normalized by the instantaneous JPL barycentric radius, the Sun's small barycentric displacement ($\sim 10^6$ km vs. $\sim 10^9$ km for the outer planets) can inflate the reported percentage. This effect is amplified further because the reported value is the maximum over all epochs, so a single small-denominator epoch can dominate the result. Absolute-error metrics clarify the situation: the Sun's maximum absolute position error is only 238 km, with RMS absolute error 159 km, both smaller than those of any other body in the full 35-body comparison. Its RMS relative error is 0.049%, which is modest and far more representative of its overall agreement with JPL. The Sun is therefore excluded from the summary mean reported in Tables 1 and 3.

Callisto overtakes Ganymede at 249 years. Callisto's error grows from 0.188% to 0.411%, surpassing Ganymede (0.290% at both durations). This is consistent with the longer orbital period of Callisto (16.7 d vs. 7.15 d), which couples to slightly different phase drift accumulation rates over multi-century timescales. Hyperion (0.226%) also emerges as a notable outlier among Saturnian moons, likely reflecting its sensitivity to perturbations from Titan and Saturn's other satellites, combined with its eccentric and inclined orbit.

Absolute position error. Table 5 supplements the relative metric with absolute position error in km for selected bodies in the 249-year simulation. This provides a normalization-free assessment of positional disagreement that is not distorted by the small barycentric radius of the Sun.

Table 5: Absolute position error (km) for the 249-year simulation, selected bodies.

Body	**Max (km)**	**RMS (km)**
Sun and Planets		
Sun	238	159
Mercury	194,898	86,793
Venus	59,238	33,903
Earth	46,554	26,303
Mars	29,946	15,532
Jupiter	16,180	9,026
Saturn	9,627	5,340
Selected Moons		
Callisto	3,111,029	1,912,082
Ganymede	2,153,442	1,504,302
Titan	2,494,262	1,903,692
Io	858,963	597,530
Moon	65,996	28,728

The Sun's maximum absolute error of 238 km over 249 years is the smallest of any body in the full 35-body comparison (Table 5 shows a representative subset), confirming that the 1.34% relative error is entirely a normalization artifact. Among planets, the error hierarchy tracks the expected importance of missing physics: Mercury (relativistic precession) > Venus and Earth (asteroid perturbations, J_2) > outer giants (minimal missing effects). The large absolute errors for Galilean moons and Titan reflect accumulated phase drift over tens of thousands of fast orbits, not growing instability.

5.3. Sources of Residual Error

The residual errors are dominated by physics model differences between the simulator (Newtonian gravity,

35 bodies) and JPL DE441. The principal missing contributions are described below.

Post-Newtonian relativity. The Einstein–Infeld–Hoffmann (EIH) equations contribute $\sim 43\,\mathrm{arcsec\,century}^{-1}$ of precession to Mercury's orbit [5]. This is the dominant error source for Mercury (0.17% at 100 yr, 0.42% at 249 yr), as the Newtonian model cannot capture this effect. Implementation of 1PN corrections was attempted; however, the velocity-dependent post-Newtonian potential breaks the separability assumption ($H = T(\mathbf{p}) + V(\mathbf{q})$) required by the Yoshida integrator. This degraded energy conservation by ~ 4 orders of magnitude (from 10^{-13} to 10^{-9}), producing worse overall accuracy than the purely Newtonian model. Implicit symplectic splitting schemes [6] or mixed-variable symplectic methods [5] offer potential paths to incorporating these corrections without sacrificing symplecticity.

Solar oblateness (J_2). The Sun's quadrupole moment produces a small additional precession, particularly affecting Mercury's perihelion advance.

Asteroid perturbations. JPL DE441 includes gravitational contributions from over 300 asteroids (principally Ceres, Pallas, and Vesta). These bodies are absent from the 35-body model.

Moon phase drift. Satellites with short orbital periods accumulate phase errors over many orbits. Io orbits Jupiter every 1.77 d and completes $\sim 20{,}600$ orbits in 100 years. At $\Delta t = 360\,\mathrm{s}$, each orbit is resolved by ~ 425 timesteps. This is adequate for 4th-order per-orbit accuracy, but small per-orbit phase errors compound over thousands of orbits. Reducing Δt or implementing adaptive timestepping would improve satellite accuracy at the cost of runtime.

Tidal and non-gravitational effects. Tidal dissipation, solar radiation pressure, and planetary ring interactions are included in DE441 but absent from the simulator.

5.4. Timestep Convergence Study

To distinguish integration error from physics model error, a systematic convergence study was performed over a range of timesteps spanning two orders of magnitude. Table 6 presents the peak-to-peak energy drift $\Delta E/E$ for the 249-year simulation at each Δt.

Table 6: Energy conservation vs. timestep (249-year simulation, 35 bodies, Yoshida 4th-order).

Δt (s)	$\Delta E/E$
60	3.74×10^{-12}
180	3.20×10^{-12}
360	1.46×10^{-12}
600	1.22×10^{-12}
900	2.08×10^{-12}
1800	2.40×10^{-11}
3600	4.79×10^{-10}
7200	4.34×10^{-8}
9600	7.55×10^{-9}

The data reveal three distinct regimes of integrator behavior.

Regime A: Precision floor ($\Delta t \leq 900\,\mathrm{s}$). For Δt from 60 s to 900 s, the energy drift remains at $\sim 10^{-12}$ with no clear 4th-order scaling. The minimum occurs near $\Delta t = 600\,\mathrm{s}$ (1.22×10^{-12}); at smaller timesteps, the drift increases mildly (to 3.74×10^{-12} at $\Delta t = 60\,\mathrm{s}$), consistent with accumulated floating-point roundoff over a larger number of arithmetic operations. This floor arises from finite-precision arithmetic in the force summation and long-time energy evaluation, not from the integrator itself. The production timestep $\Delta t = 360\,\mathrm{s}$ is well within this converged regime.

Regime B: 4th-order convergence ($900 \leq \Delta t \leq 3600\,\mathrm{s}$). In this range, the energy error scales approximately as $\mathcal{O}(\Delta t^4)$, consistent with the expected behavior of the Yoshida integrator. From $\Delta t = 900\,\mathrm{s}$ to 1800 s (a factor of 2), the energy drift increases by a factor of ~ 11.5 (expected: $2^4 = 16$). From 1800 s to 3600 s, the drift increases by a factor of ~ 20 (expected: 16). Both ratios are in reasonable agreement with 4th-order scaling; the deviation from the theoretical factor of 16 is expected when measur-

ing peak-to-peak variation over a long integration rather than single-step local truncation error.

Regime C: Dynamical under-resolution ($\Delta t \geq 7200\,\mathrm{s}$). At $\Delta t = 7200\,\mathrm{s}$ ($\sim 2\,\mathrm{hr}$), the energy drift exceeds the 4th-order prediction by a large factor: the ratio from 3600 s to 7200 s is ~ 90, far exceeding the expected 16. At this timestep, the fastest-orbiting body (Io, period 1.77 d) is resolved by only ~ 21 timesteps per orbit. The symplectic splitting approximation degrades when the timestep becomes comparable to the dynamical timescale of the system. The non-monotonic behavior at $\Delta t = 9600\,\mathrm{s}$ ($\Delta E/E$ decreasing relative to 7200 s) is consistent with the oscillatory energy error characteristic of symplectic integrators: the peak-to-peak variation depends on the resonance between the timestep and the orbital frequencies, and can fluctuate non-monotonically at large Δt [2].

Implications. The convergence study strongly indicates that residual positional errors are dominated by model discrepancy rather than integrator truncation error. In Regime A, the integration error has converged to the floating-point floor, yet positional errors against JPL remain unchanged. Furthermore, the error hierarchy matches the expected importance of missing physics: Mercury (relativity) > Galilean moons (phase drift) > outer planets (minimal missing effects). Any timestep within Regime A ($\Delta t \lesssim 900\,\mathrm{s}$) produces indistinguishable energy conservation; the production timestep $\Delta t = 360\,\mathrm{s}$ is adopted for a modest additional margin.

6. Performance

6.1. Computational Complexity

The force computation scales as $\mathcal{O}(N^2)$ per timestep due to direct pairwise summation. The Yoshida integrator requires 3 force evaluations per step, yielding a total cost of $\mathcal{O}(3N^2 \cdot T/\Delta t)$ for a simulation of duration T.

For the 35-body solar system with $\Delta t = 360\,\mathrm{s}$ over 249 years (adopting a Julian year of 365.25 d), this corresponds to $3 \times 35^2 \times 21{,}827{,}340 \approx 8.0 \times 10^{10}$ pairwise interaction evaluations, completed in $\sim 172\,\mathrm{s}$ on a single core.

6.2. Parallelization

The outer loop of the force computation is parallelized with OpenMP for particle counts exceeding a configurable threshold ($N \geq 500$). The inner loop uses `#pragma omp simd` with reductions to enable SIMD auto-vectorization. For the 35-body solar system, the overhead of thread management exceeds the parallelization benefit, so OpenMP threading is automatically disabled. SIMD vectorization of the inner loop remains active regardless of N.

A scaling benchmark sweeps N from 512 to 131,072, auto-calibrating step count per configuration and reporting median wall time over multiple trials. At $N = 131{,}072$ on a 6-core desktop CPU (AMD Ryzen 5 8500G), OpenMP yields 4.60× speedup (124,810 ms serial to 27,146 ms parallel), with estimated throughput reaching $\sim 154\,\mathrm{GFLOP/s}$ compared with $\sim 33\,\mathrm{GFLOP/s}$ for the serial baseline. Scaling improves with N and then saturates, suggesting limitation by shared hardware resources such as memory bandwidth and cache hierarchy.

6.3. Memory Efficiency

The SoA layout stores 13 scalar fields per particle (3 positions, 3 velocities, 3 current accelerations, 3 old accelerations for Velocity Verlet, and mass). All sub-arrays occupy a single contiguous allocation with each sub-array start aligned to the SIMD register width (32 bytes for AVX2, 64 bytes for AVX-512). Sub-arrays are padded to a SIMD-element boundary, so the total allocation is $13 \times \lceil N/w \rceil \times w$ doubles, where w is the SIMD width in elements. For $N = 35$, the entire particle dataset occupies $\sim 3.6\,\mathrm{KB}$, fitting comfortably within L1 cache (typically 32–64 KB). The `__restrict__` qualifiers permit the compiler to assume no pointer aliasing, enabling store-to-load forwarding optimizations.

7. Conclusion

We have presented a symplectic N-body simulator that achieves near-machine-precision energy conservation ($\Delta E/E \leq 10^{-12}$) and angular momentum conservation ($\Delta L/L \leq 10^{-13}$) over multi-century integration of the solar system. The Yoshida 4th-order integrator provides an excellent balance between accuracy and computational cost, requiring only 3 force evaluations per timestep while delivering 4th-order convergence and bounded energy error.

Positional accuracy against JPL Horizons DE441 is 0.079% mean maximum relative error over 249 years across 34 non-Sun bodies. Supplemental absolute-error analysis shows that the Sun's maximum absolute position error is only 238 km, despite its inflated barycentric relative error, confirming that the Sun outlier is a normalization artifact rather than a dynamical failure. The dominant residual errors remain consistent with missing physics in the Newtonian 35-body model, particularly relativistic precession for Mercury and long-horizon phase drift for fast-orbiting satellites.

A systematic timestep convergence study identifies $\Delta t \lesssim 900\,\mathrm{s}$ as the converged regime where the integration error has reached the floating-point precision floor. Below this threshold, no further improvement in energy conservation is achievable with double-precision arithmetic. The production runs use $\Delta t = 360\,\mathrm{s}$.

Future work includes implementing post-Newtonian corrections via an implicit or mixed-variable symplectic scheme, adding solar oblateness perturbations, and incorporating adaptive timestepping for improved resolution of fast-orbiting satellites.

Code Availability

The source code for the simulator and the validation pipeline are open source and available at https://github.com/karl-kes/N-Body-Physics-Engine.

References

[1] H. Yoshida, "Construction of higher order symplectic integrators," *Phys. Lett. A*, vol. 150, no. 5–7, pp. 262–268, 1990.

[2] E. Hairer, C. Lubich, and G. Wanner, *Geometric Numerical Integration: Structure-Preserving Algorithms for Ordinary Differential Equations*, 2nd ed. Berlin: Springer, 2006.

[3] R. S. Park *et al.*, "The JPL planetary and lunar ephemerides DE440 and DE441," *Astron. J.*, vol. 161, no. 3, p. 105, 2021.

[4] L. Verlet, "Computer 'experiments' on classical fluids. I. Thermodynamical properties of Lennard-Jones molecules," *Phys. Rev.*, vol. 159, no. 1, pp. 98–103, 1967.

[5] J. Wisdom and M. Holman, "Symplectic maps for the N-body problem," *Astron. J.*, vol. 102, pp. 1528–1538, 1991.

[6] J. E. Chambers, "A hybrid symplectic integrator that permits close encounters between massive bodies," *Mon. Not. R. Astron. Soc.*, vol. 304, no. 4, pp. 793–799, 1999.

[7] H. Rein and D. S. Spiegel, "IAS15: a fast, adaptive, high-order integrator for gravitational dynamics, accurate to machine precision over a billion orbits," *Mon. Not. R. Astron. Soc.*, vol. 446, no. 2, pp. 1424–1437, 2015.